

Self-Styled Neural Painter

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Figure 1. Several art styles of paintings created by Self-Styled Neural Painter based on the input images. The second image from Pixabay user UNTHINKER.

ABSTRACT

This work introduces Self-Styled Neural Painter (SSNP) creating stylized artworks in a stroke-by-stroke manner. SSNP consists of digit artist, canvas, style-stroke generator (SSG). By using SSG to generate style strokes, SSNP creates different styles paintings based on the given images. We design SSG as a three-player game based on a generative adversarial network to produce pure-color strokes that are crucial for mimicking the physical strokes. Furthermore, the digital artist adjusts parameters of strokes (shape, size, transparency, and color) to reconstruct as much detailed content of the reference image as possible to improve the fidelity.

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1 INTRODUCTION

Teach a machine mimicking human artists needs to solve three critical issues: 1) allowing the computer paint on the canvas in a stroke-by-stroke manner with a certain order; 2) creating strokes containing stroke textures similar to physical strokes of human artists; 3) preserving detailed content of the reference image like a real painting. Conventional methods such as the stroke-based rendering (SBR) method [Hertzmann 1998] contribute to stroke modeling. Learning-based methods [Huang et al. 2019; Nakano 2019; Zou et al. 2021] have flexible frameworks adaptive to diverse

art styles. Although these methods generate painting-like images, they still lose content details and have limited art styles of paintings.

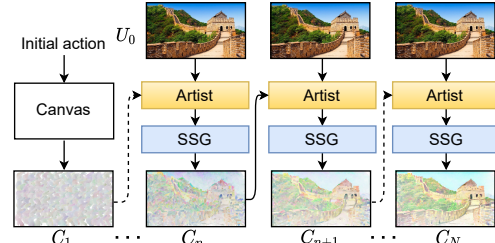


Figure 2. Learning-to-paint process. The terms of U_0 , C_n and N denote the reference image, a certain state of the canvas, and the painting times, respectively. The artist optimizes the stroke generated by SSG to continually paint.

To address the above limitations, we propose a new method namely *Self-stylized Neural Painter* (SSNP), which leverages advantages of both SBR methods and learning-based methods. We first design a style-stroke generator (SSG) to produce diverse styles of strokes with adjustable stroke shapes, size, transparency, and color. We then design an *artist* module optimizing strokes generated by SSG to create styles of paintings, such as oil paintings, watercolor paintings, pastel paintings, and ink-wash paintings (as shown in Figure 1). The process of SSNP painting imitates the learning process of a novice painter to a veteran, as shown in Figure 2. We conduct extensive experiments to evaluate SSNP. Experimental results demonstrate that SSNP generates different styles of paintings, which also contain sufficient-detailed contents (e.g., face textures and details of buildings) and have a higher similarity to given images while retaining clear stroke textures. Meanwhile, a user study also demonstrates that images generated by our SSNP obtain higher scores than those generated by other methods.

2 TECHNICAL APPROACH

Our SSNP mainly consists of the digital artist, the canvas, and SSG. We design SSG based on a generative adversarial network

(GAN) to generate pure-color strokes. We design the artist module to optimize parameters of strokes generated by SSG, thereby rendering the optimized strokes onto the canvas. The initial strokes are randomly generated to cover the entire canvas and the parameters are optimized to reconstruct the input image. The parameters control the stroke shape, size, transparency, and color instead of the styles and textures of the stroke. We artfully orchestrate SSG and the digital artist to recreate paintings with different art styles.

2.1 Stroke datasets

The stroke style affects the art style of a painting. We devise a stroke designer to generate stroke datasets for training SSG. We mainly adopt four most representative stroke styles (though more than five styles of strokes are obtained): 1) watercolor style, 2) oil-painting style, 3) ink-wash painting style, and 4) pastel-painting style. We use Bézier curves to simulate the route of a stroke and variant circles to mimic the tip of a brush. The first three styles of strokes are generated by the stroke designer while the pastel-painting style is provided by [Nakano 2019].

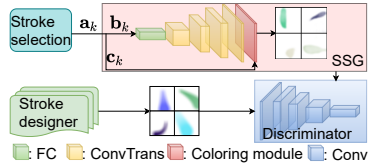


Figure 3. Training process of SSG.

2.2 Style-stroke generator

We design SSG mainly based on Deep Convolutional Generative Adversarial Network (DCGAN) [Radford et al. 2016]. However, strokes generated by DCGAN are stained with shade colors. It is difficult to mimic human painting process by using stained-colored strokes. To control stroke colors, we design the coloring module as a third player (i.e., the red cuboid) immediately after the original generator as shown in Figure 3. Stroke Selection is controlled by a set of parameters $\mathbf{a}_k$ (used as inputs of SSG), where $\mathbf{a}_k = [\mathbf{b}_k, \mathbf{c}_k]$ ($k = 1, \dots, K$ and K is the batch size). The set $\mathbf{c}_k$ controls the stroke color, and $\mathbf{b}_k$ controls other characters of the stroke. Since SSG learns the color of the reference image U_0 directly by feeding in $\mathbf{c}_k$, SSG endows the unified framework with a high fidelity of details.

2.3 Digital artist

It is crucial to optimize the stroke selection. The artist first extracts features from both the reference image U_0 and the state of the canvas C_n and then computes the distance between U_0 and C_n by the ℓ_1 -distance. The stroke-selection-optimization (SSO) algorithm optimizes the stroke selection by tuning the parameters in each $\mathbf{a}_k$ according to the ℓ_1 -distance loss function. The SSO process is achieved by the back-propagation of ℓ_1 -distance loss function with respect to each weight by the chain rule. Each C_n is composed of K strokes, and each stroke is determined by $\mathbf{a}_k$. Thus, the values of the parameters in each $\mathbf{a}_k$ are optimized by the back-propagation of ℓ_1 -distance loss function.

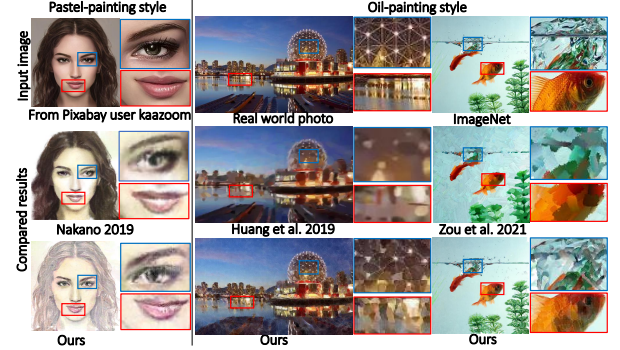


Figure 4. Visual comparisons between ours and SOTA methods. The image of the girl is from Pixabay user kaazoom.

3 EVALUATION

We evaluate SSNP by images covering portraits, animals, landscapes, and buildings. Figure 4 shows the visual comparison between our method and state-of-the-art (SOTA) methods. Our paintings with different art styles contain more detailed content than others. To further evaluate our model and SOTA methods, we conduct a two-step user study: user study I and user study II. For fair comparison, we conduct blind trials, where neither participants knew the SOTA methods nor our SSNP. We conduct two questionnaires for pastel-painting style and oil-painting style. User study I focuses on the recognizability of the art-style. User study II focuses on the fidelity of the content details and the recognizability of stroke textures. Table 1 shows the results of user study II, where the scores of the content details and the recognizability of stroke textures of our paintings are higher than the compared methods.

Table 1: User study of content details and stroke textures.

Items	Methods	μ	σ	95% confidence interval	
				Lower Bound	Upper Bound
Content	Nakano [Nakano 2019]	3.505	0.242	3.403	3.607
	Ours	3.690	0.176	3.616	3.764
	Huang et al. [Huang et al. 2019]	2.735	0.131	2.68	2.79
	Ours	3.549	0.248	3.445	3.654
Stroke	Nakano [Nakano 2019]	3.603	0.284	3.483	3.722
	Ours	3.737	0.218	3.645	3.829
	Huang et al. [Huang et al. 2019]	2.691	0.133	2.635	2.747
	Ours	3.473	0.213	3.383	3.562

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